BATCHAYA YACYNTE

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SUMMARY

Robotics and Autonomous Systems Engineer specializing in real-time perception, multi-sensor state estimation, and embedded AI systems. Proven experience developing high-performance C++ and Python software for UAVs and autonomous robotic platforms on resource-constrained hardware, including NVIDIA Jetson and ARM-based systems. Skilled in visual odometry, probabilistic tracking, closed-loop visual control, and translating advanced robotics theory into resilient, low-latency production software.

TECHNICAL SKILLS

- Estimation & Localization: Monocular & Stereo Visual-Inertial Odometry (VIO), Multi-Sensor Fusion (Stereo Camera/LiDAR/IMU), Extended Kalman Filters (EKF), 3D Point-Cloud Surface Reconstruction, State Estimation under uncertainty.
- Computer Vision & Neuromorphic Perception: Event-Based Vision (DVS), Asynchronous Optical Flow, Outlier Rejection (RANSAC, Affine Refinement), SIFT & Lucas-Kanade Feature Tracking, Epipolar Geometry, Camera & Hand-Eye Calibration.
- Software Engineering: Modern C++ (Object-Oriented, low-latency optimization), Python, ROS / ROS2, Linux (Embedded Kernel/OS Tuning, Shared Memory & GStreamer Pipeline Deployment), Git.
- Hardware & Embeddings: Nvidia Jetson Developer Kits, Rapsberry Pi, Radxa, Pixhawk/Drone Flight Controllers, ARM Cortex (Infineon XMC), Motorized Industrial Lenses.

PROFESSIONAL EXPERIENCE

Almotion Bavaria | Graduate Research Assistant | Ingolstadt, Germany **Apr 2025 - Present**

- Core Technical Lead for Project ENGEL: Designing and deploying an end-to-end custom monocular visual odometry pipeline explicitly optimized for autonomous UAV repositioning, stabilization, and visual servoing.
- Arbitrary 3D Surface Modeling: Eliminated standard flat-plane homography assumptions by architecting a geometric backend utilizing SIFT feature matching and essential matrix decomposition, enabling accurate continuous ego-motion estimation over highly non-planar topographies.
- Closed-Loop Flight Control: Engineered a robust flight-correction loop by tightly binding the real-time visual state estimation variables into a custom sigmoid-based control loop, securing aggressive and smooth quadcopter positioning.

Fraunhofer IVI - Vehicle Technologies | Graduate Research Assistant **Apr 2025 - Mar 2026**

- Probabilistic Optical Flow Architecture: Developed a novel probabilistic matcher extension for a triplet-based event optical flow framework. Implemented a Gaussian residual model with motion-adaptive variance scaling to quantify temporal uncertainty and improve robustness against asynchronous sensor noise.
- Real-Time C++ Optimization: Engineered a highly optimized single-threaded Modern C++ pipeline capable of processing raw neuromorphic streams at 3.5 million events per second (MEPS) on embedded hardware, avoiding multithreading overhead while maintaining deterministic latency.
- Geometric Validation & Embedded Deployment: Designed an affine RANSAC refinement stage to reject spurious flow vectors and maximize inlier consistency. Deployed the full Linux and ROS/ROS2 acquisition stack on NVIDIA Jetson hardware for synchronized capture from an IDS USB 3 uEye XCP-E event camera and evaluation against the MVSEC dataset.

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PROFESSIONAL EXPERIENCE

Agrobot | Robotics Engineer | Yaoundé, Cameroon (Remote) **Apr 2024 - Apr 2025**

- Multi-Sensor Fusion: Designed and deployed a real-time state estimation framework integrating stereo vision, 3D LiDAR, and high-frequency IMU data for autonomous robotic navigation.
- State Estimation & Localization: Developed an Extended Kalman Filter (EKF) backend supporting tightly and loosely coupled sensor fusion, significantly reducing localization drift during field operations.
- 3D Perception & Terrain Mapping: Built a real-time 3D reconstruction and mapping pipeline combining dense stereo and LiDAR point clouds for autonomous navigation across uneven terrain and topographic analysis.

IDEE | Bachelor’s Thesis & REM 2024 Publication | IEEE REM 2024 **Nov 2023 - Mar 2024**

- Developed a monocular visual-inertial odometry (VIO) framework designed to reduce catastrophic localization drift during aggressive UAV motion.
- Engineered a geometric outlier rejection pipeline using epipolar constraints to eliminate mathematically inconsistent feature correspondences from Lucas–Kanade optical flow tracks.
- Validated the system across Unreal Engine simulation environments, real-world DJI Tello flight experiments, and the KITTI odometry benchmark dataset.
- Research work was peer-reviewed and published in the proceedings of the REM 2024 conference in the IEEE Xplore digital library.

Institut Digital Engineering (IDEE) | Robotics Intern | Schweinfurt **Oct 2022 - Nov 2023**

- Developed and validated core computer vision pipelines, including optical flow and time to collision for robotic perception and localization.
- Programmed autonomous flight control profiles for physical UAV platforms and designed ROS-based validation workflows for autonomous navigation testing.
- Engineered photorealistic, physics-based simulation environments using ROS and Unreal Engine to evaluate localization and control algorithms before hardware deployment.
- Modeled and analyzed multi-modal sensor noise characteristics for LiDAR tracking systems and wheel encoder telemetry to improve estimation reliability.
- Designed high-fidelity CAD assemblies and production layout configurations for an automated vehicle assembly line.

EDUCATION

- **M.Eng. in AI Engineering and Autonomous Systems | GPA: 2.5 (DE scale)**
  - Technische Hochschule Ingolstadt, Germany: March 2025 to August 2026 (Expected)
- **B.Eng. in Mechatronics Engineering | GPA: 2.3 (DE scale)**
  - Technische Hochschule Würzburg-Schweinfurt, Germany: March 2020 to August 2024

PUBLICATION

Batchaya, Y. D., et al. “Getting Started With a Simple Visual-Inertial Odometry” REM 2024 (IEEE)
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Languages: English, French, German (B1/B2).